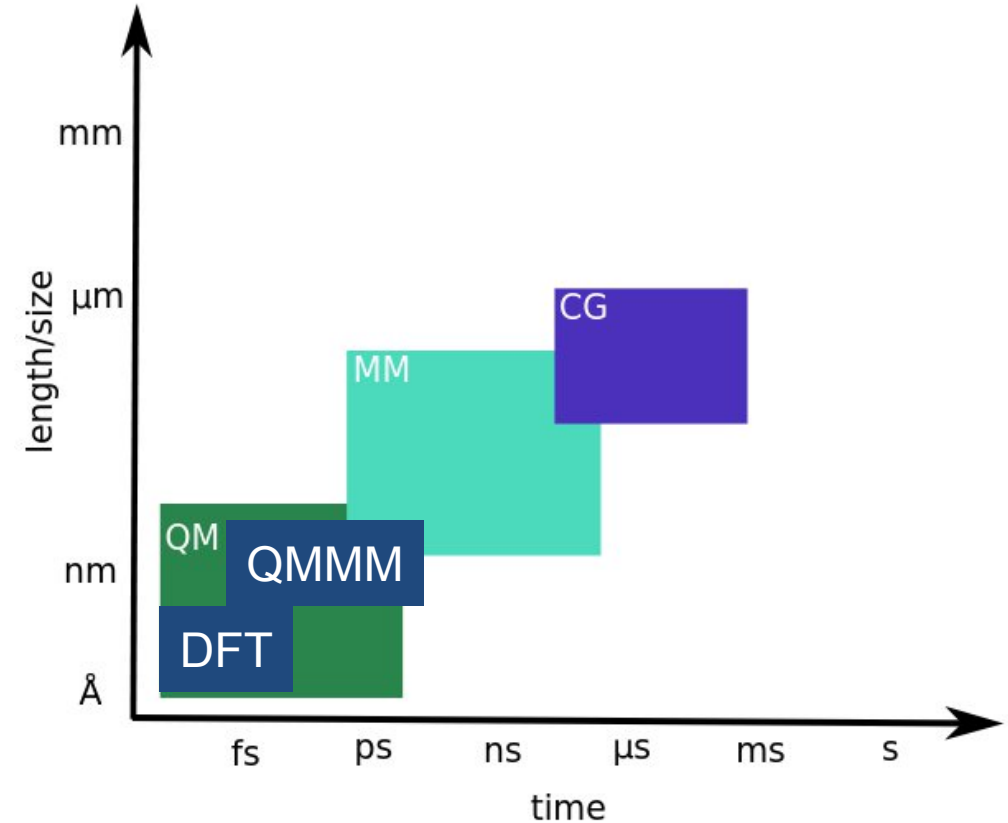


Time vs Length Scale

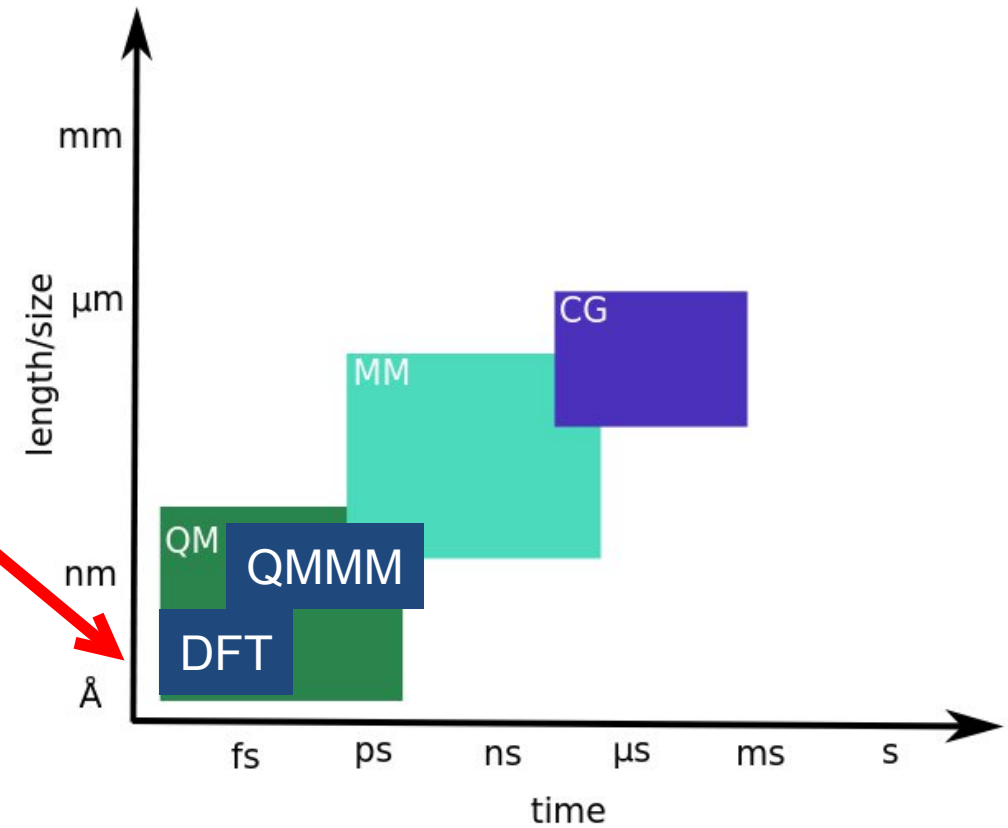
Time vs Length Scale

- hierarchy of theories available
- with accuracy computational cost increases
- limits system size and simulation duration
- a wide range of approximations



Time vs Length. Adopted from [1]

DFT: Density Functional Theory

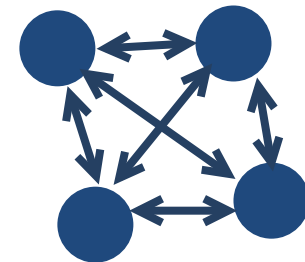


Time vs Length. Adopted from [1]

Density Functional Theory

DFT

- ground state electron density($\rho(\mathbf{r})$) alone determines the system behavior
- overcomes many-body problem



CP2K - Quickstep Method - DFT GPW [1]

- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$E[\rho(\mathbf{r})] = E^T[\chi(\mathbf{r})] + E^{\text{ne}}[\rho(\mathbf{r})] + E^{\text{ee}}[\rho(\mathbf{r})] + E^{\text{nn}} + E^{\text{XC}}[\rho(\mathbf{r})]$$

K.E electrons	nuclear-electron	electron-electron	nuclear-nuclear	XC
Gaussian	Gaussian+PW	PW	PW	PW

CP2K InputFile

$$E[\rho(\mathbf{r})] = E^T[\chi(\mathbf{r})] + E^{\text{ne}}[\rho(\mathbf{r})] + E^{\text{ee}}[\rho(\mathbf{r})] + E^{\text{nn}} + E^{\text{XC}}[\rho(\mathbf{r})]$$



```
&FORCE_EVAL
  &DFT
    CHARGE 0
    &MGRID
      NGRIDS 4
      REL_CUTOFF 30
    &END MGRID
    &QS
      METHOD GPW
      EPS_DEFAULT 1.0E-12
      EXTRAPOLATION ASPC
    &END QS
  &SCF
    SCF_GUESS RESTART
    EPS_SCF 1.0E-8
    CHOLESKY INVERSE
    &DIAGONALIZATION
      ALGORITHM STANDARD
    &END DIAGONALIZATION
    &MIXING
      METHOD BROYDEN_MIXING
      NBROYDEN 16
    &END MIXING
  &END SCF
```

Grids/Mesh

QM Method and Settings

Fine tune until consistent solution is reached

```
&XC
  &XC_FUNCTIONAL PBE
  &END XC_FUNCTIONAL
  &VDW_POTENTIAL
    POTENTIAL_TYPE PAIR_POTENTIAL
    &PAIR_POTENTIAL
      PARAMETER_FILE_NAME dftd3.dat
      TYPE DFTD3
      CALCULATE_C9_TERM .TRUE.
      REFERENCE_C9_TERM .TRUE.
      REFERENCE_FUNCTIONAL PBE
    &END PAIR_POTENTIAL
  &END VDW_POTENTIAL
&END XC
&END DFT
&END FORCE_EVAL
```

VdW and XC

```
&CELL      cell dimensions
  PERIODIC XYZ
  A 17.8    0.    0.
  B 0.      19.15 0.
  C 0.      0.    23.21
&END CELL
```

```
&KIND 0
  BASIS_SET SZV-MOLOPT-GTH
  POTENTIAL GTH-PBE
&END KIND
&KIND H
  BASIS_SET SZV-MOLOPT-GTH
  POTENTIAL GTH-PBE
&END KIND
```

Basis Sets and N

```
calc = CP2K(label='benzene',
            cutoff=250,
            basis_set='SZV-MOLOPT-GTH',
            pseudo_potential='auto',
            charge=0,
            xc='PBE',
            print_level='LOW',
            max_scf=20,
            inp=inp)
```

- project name
- electronic cutoff
- electronic wavefunction
- nuclear
- total System Charge
- exchange correlation
- output setting
- max number of SCF cycles

Example problem overview

Importance of Phosphorus

Essential for plant and animal life



Part of DNA, RNA



Brain functioning



Oxygen delivery



Photosynthesis [1]



Improve flowering



Root development

[1] Plants 12(15), 2861 (2023)

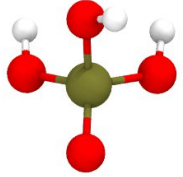
[2] Plant and Soil 348, 29 (2011)

Limiting nutrient for plant growth

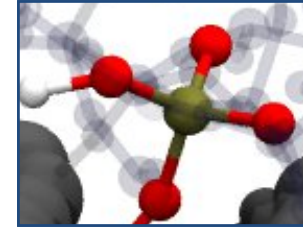
- Only small amount of P in soil
- Unlike C, N, O, H: P no gaseous phase
- Forests evolved strategies [2]
- Used in fertilizers: global food security



Example problem overview



Phosphoric acid (H_3PO_4)



Phosphate (HPO_4^{-2}) [2]

[2] Soil Sci. Plant Anal. 11, 835–842 (1980)

AIM

BioPhyCourse/Day2/03_CP2K_QMMM/data/CP2K-DFT

CP2K-Intro: Here we optimize Phosphoric acid using CP2K inputfile

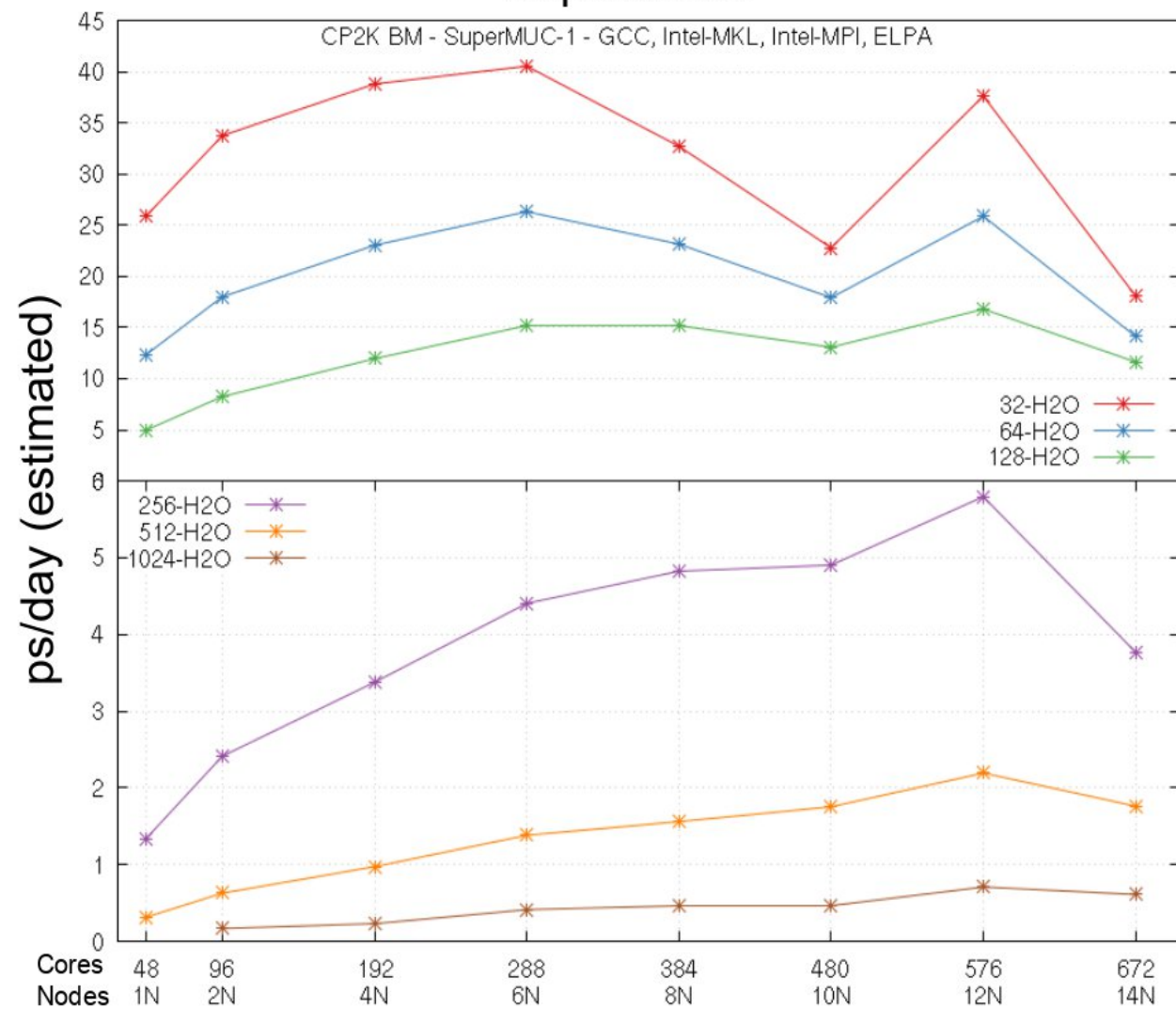
CP2K-ASE: Here we solvate Phosphoric acid to see if it deprotonates using ASE+CP2K

Benchmarks

1picosecond simulation of
1000 water molecules/ 3000 atoms
in 1 and half day.



SuperMUC1



SuperMUC2

